

Preferential Leaching in Large Undisturbed Soil Blocks from Conventional Tillage and No-Till Fields in Southern Alberta

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There is a concern that adoption of conservation tillage practices such as no-till may increase preferential leaching of water and chemicals to the groundwater. Evidence from previous studies of long-term (since 1968) conventional tillage (CT) and no-till (NT) fields in southern Alberta suggests that tillage practice has the potential to influence preferential leaching. However, no studies have been done to test this hypothesis. Our objective was to utilize the two-region (mobile/immobile water) model of solute transport to compare preferential leaching in CT and NT fields of clay loam texture. Four large ($46 \times 46 \times 51$ cm) undisturbed blocks of soil were excavated from each tillage field (unreplicated) during the fallow phase of a wheat-fallow rotation. The soil blocks were transported to the laboratory, stored under drying conditions (32 months), and miscible displacement experiments conducted under saturated, steady-state conditions. Because of the long storage period, the focus of our study was on long-term (e.g., structure, earthworm burrows, old root channels) rather than short-term (e.g., tillage of surface soil) tillage effects on preferential leaching. Breakthrough curves (BTCs) for chloride were derived, and the modified convection-dispersion equation fitted to the experimental data using a two-region model (CXTFIT-Version 2.0) to allow estimation of the mobile water fraction (β) or extent of preferential leaching. Breakthrough curves for both tillage fields exhibited early initial breakthrough, a rapid rise in tracer concentration, a shift of the BTC peak to the left of one pore volume, and a slow decline ("tailing") toward zero concentration for the descending limb of the curve. Reasonably good fits were obtained for fitting of the two-region model to the BTCs, as indicated by correlation coefficients ranging from 0.60 to 0.84. Mean values for the mobile water fraction were similar for the CT field (0.78) and NT field (0.80), suggesting no difference in preferential leaching of chloride. We hypothesize that the extent of preferential leaching in earthworm burrows (*Aporrectodea caliginosa*) in the NT field and leaching in old root channels in the CT field may have been similar. Further research on replicated plots is needed to examine short-term effects of tillage practice (e.g., tillage of surface soil) as well as the individual contribution of earthworm burrows and old root channels, to preferential leaching under CT and NT.

Key words: miscible displacement, breakthrough curves, preferential leaching, tillage system

Introduction

Conservation tillage, which decreases or eliminates tillage operations and maintains greater amounts of crop residue on the soil surface, is

increasing on the Canadian prairies (Larney et al. 1994). Although the benefits associated with conservation tillage include improved soil and water conservation and reduced fossil fuel inputs, there is a concern about greater potential leaching of water, nutrients and pesticides to the groundwater. Greater potential leaching for no-till (NT) has been attributed to decreased runoff and increased infiltration resulting from an increase in surface residues, a greater bioporosity caused mainly by earthworm burrows, and soil pores that are more continuous because they are not disrupted by tillage (Baker 1987; Dick and Daniel 1987; Logan et al. 1991). Consequently, the relative impact of conservation tillage versus conventional tillage (CT) practices on preferential solute leaching and the potential effect on groundwater quality warrants investigation.

Preferential flow (also known as bypass or channelling flow) has been defined by Bevan (1991). During wetting, local wetting fronts propagate into the soil to significant depths, thus bypassing intervening matrix pore space. The result is movement of water and solutes to far greater depths at a much faster rate than predicted by Richards flow equation. Preferential flow may occur in non-capillary structural voids (cracks, root channels, worm channels, etc.), through zones of locally high conductivity in capillary sized pores, and even in soils that show no obvious structural channels or in packed laboratory columns (Bevan 1981).

Breakthrough curves (BTCs) derived from miscible displacement experiments are extremely useful for estimating preferential leaching in macropores (Busseau and Rao 1990). Experimental BTCs can be fitted using computer models and least-squares optimization techniques using one-region or two-region models (van Genuchten 1981; Parker and van Genuchten 1984; Toride et al. 1995). One-region models utilize the convection-dispersion equation (CDE) for describing transport of solutes through homogeneous soil. However, this simple equation often fails to provide a satisfactory description of solute transport in undisturbed soils containing macropores. Therefore, two-region models have been developed to account for preferential flow. The CDE has been modified so the mobile water fraction (preferential flow) and immobile water fraction can be estimated from fitting of the modified CDE to experimental BTCs.

Miscible displacement experiments have been conducted on undisturbed soil columns or soil blocks with intact structure (Anderson and Bouma 1977; Bowman et al. 1994; Wildenschild et al. 1994; Phillips et al. 1995; Jensen et al. 1996). However, few studies have compared BTCs and fitted parameters of solute transport models under different management practices such as tillage (Andreini and Steenhuis 1990; Singh and Kanwar 1991). Andreini and Steenhuis (1990) determined the breakthrough of bromide and a blue dye on large intact soil blocks excavated from CT and NT (since 1970) corn fields (fine sandy loam) in New York. They conducted their breakthrough experiments under unsaturated and steady-state conditions. Breakthrough curves indicated preferential leaching in each tillage treatment; however, solute transport parameters were similar for CT and NT. They attributed the lack of a tillage effect to the low flow rate

used in the experiment and the timing of their sampling in relation to the most recent tillage event. Singh and Kanwar (1991) excavated large soil columns from CT and NT corn fields (loam) in Iowa that had been no-tilled for 6 years. Their breakthrough curves exhibited preferential leaching in both tillage treatments, with a higher degree of preferential flow in NT than CT.

The two previous miscible displacement-tillage studies were both conducted in the eastern region of North America. We are unaware of similar studies in the Northern Great Plains Region of North America. In addition, these studies both used the one-region model for describing solute transport. We are unaware of any studies that have estimated preferential flow by fitting the two-region model to BTCs derived from undisturbed soil from long-term CT and NT fields.

A long-term (since 1968) study was initiated at Lethbridge to test CT and NT cropping systems. Although the CT and NT fields are not replicated, the no-till plots are among the oldest in North America. Previous studies have reported differences in rooting density and soil porosity (Volkmar and Entz 1995; Miller et al. 1998), soil hydraulic properties (Miller et al. 1999) and earthworm populations (Clapperton et al. 1997) between CT and NT fields.

Previous study of these long-term tillage fields found a similar soil macro-structure in the CT and NT fields, which suggested little potential effect of soil structure on preferential leaching. In contrast, rooting density was found to be higher in the CT field than the NT field, and was attributed to the greater penetration resistance and compaction under NT (Volkmar and Entz 1995). Greater rooting density under CT, therefore, has the potential to increase preferential leaching through old root channels. Long-term storage of previously cropped soil cores (to allow roots to decompose) is a commonly used technique to examine the effect of old root channels on solute leaching (Gish and Jury 1983; Li and Ghodrati 1994).

Tillage studies in eastern North America have found that NT fields are dominated by the "night-crawler" or earthworm species *Lumbricus terrestris* (Edwards et al. 1992). This deep-burrowing species lives in nearly vertical burrows, often 5 mm in diameter or greater, and 1 to 3 m deep, depending on soil and environmental conditions (Lee 1985). These large burrows or macropores can contribute to considerable leaching of water and agrochemicals through NT soils (Edwards et al. 1989). In contrast, the long-term NT field at Lethbridge is dominated by the species *Aporrectodea caliginosa* (Clapperton et al. 1997). This species tends to form shallow (<25 cm depth) and more randomly oriented burrows (McKenzie and Dexter 1993). Some studies have found that burrows of *Aporrectodea caliginosa* contribute to increased infiltration and percolation of water (Joschko et al. 1992; Trojan and Linden 1992), whereas other studies have found little or no contribution to increased infiltration (Ela et al. 1992). We are unaware of any studies in the Northern Great Plains Region that have conducted miscible displacement experiments on NT soils containing this earthworm species to ascertain their potential effect on preferential leaching.

Evidence from previous studies on these long-term tillage fields suggests that tillage practice has the potential to influence preferential leaching, but no studies have been done to test this hypothesis. Our objective was therefore to utilize the two-region model of solute transport to compare preferential leaching in an undisturbed clay loam soil from the Northern Great Plains Region of North America that had been under long-term CT and NT management. Because of the long storage period for our soil blocks, we focussed on the long-term (e.g., soil structure, rooting density, earthworm burrows) rather than short-term (e.g., tillage of surface soil) effects of tillage practice on preferential leaching.

Materials and Methods

Study Area

The tillage experiment (non-replicated) was established in 1968 on a Dark Brown Chernozemic (Typic Haploboroll) clay loam soil (Chokio series) at the Lethbridge Research Centre, Lethbridge, Alberta. Soils of the Chokio series are dominantly Calcareous Dark Brown with inclusions of Orthic Dark Brown (Alberta Institute of Pedology 1980). These soils developed on fine-loamy to fine-silty lacustrine material, and the surface expression is undulating with Class 2 slopes (0.5–2.5%). The plots are 23 m wide and 188 m long (0.4 ha). The sampled plots were in the fallow phase of a 2-year wheat-fallow rotation consisting of wheat-cultivated fallow (conventional tillage, CT) and wheat-chemical fallow (no-tillage, NT). In the cropped phase, hard red spring wheat (*Triticum aestivum* L.) was seeded using a hoe-drill. Conventional tillage was performed with a chisel plow or heavy-duty cultivator and rodweeder attachment to a depth of approximately 10 cm. During the year of sampling (1992), cultivation of CT was performed on May 7 and August 10. No-tillage (NT) used only herbicides for weed control.

Field Sampling

Sampling locations were taken along a north-south transect bisecting the western half of each tillage plot. Four sampling locations were spaced 38 m apart along the transect. Large undisturbed soil blocks were excavated from each sampling location. Soil blocks from NT were taken between June 18 and July 17, and for CT they were taken between August 31 and September 2, 1992. An undisturbed block of soil (46 x 46 x 51 cm) was carved out using a shovel, hand trowel and straight-edge guide. A plywood box (50 x 50 x 60 cm; 2 cm wall thickness) with top and bottom ends removed was inserted over the soil block, and the cavity (2 cm) between the soil and plywood was filled with an expanding polyurethane foam from a pressurized can. The soil encased in the box was detached by soaking the bottom of the block overnight and using a wire cable attached to a pulley to slice the bottom of the soil. The soil block was removed, the

bottom trimmed flush to the box, and plywood covers screwed onto the top and bottom of the box.

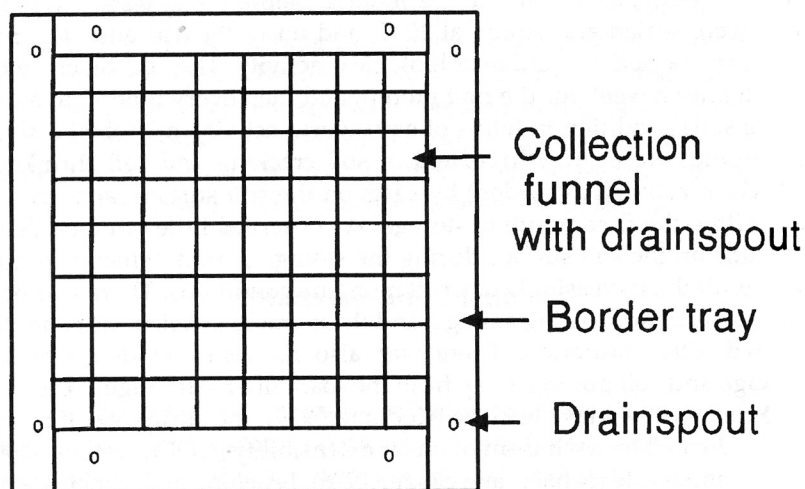
Soil blocks were transported to the laboratory at the Lethbridge Research Centre, and stored for 32 months before being tested. The soil blocks were sealed and stored at 20°C, and no water was added during the storage period to minimize biological activity. The soil blocks were checked once a week for the first month, and then every month, to ascertain the soils condition in terms of earthworm activity, microbial activity (e.g., mould, algae growth), extent of soil cracking and soil shrinkage. Earthworm activity, as evident by casts on the soil surface, appeared to cease within the first month of storage. We observed little or no evidence of moulds on the soil surface during the storage period, which was consistent with the increasingly dry soil moisture conditions. There was only minor soil cracking during storage, and these cracks swelled shut and disappeared after saturation. There was also no visual evidence of soil shrinkage and soil pulling away from the foam liner. Although these soils are dominated by smectite clays (Kodama 1979), the shrink-swell potential, as indicated by coefficient of linear extensibility (COLE) values (0.06), was only moderate (Schafer and Singer 1976). Leaching experiments were initiated in May 1995, and were completed in May 1996.

Miscible Displacement Experiment

A funnel-lysimeter (FL) used to measure preferential leaching and the spatial variation of leaching was utilized to collect effluent from the bottom of the soil block. It consisted of 64 square 5 x 5 x 4 cm funnels (stainless steel) in the centre, surrounded by two edge-trays of 5 x 40 x 4 cm and two edge-trays of 5 x 50 x 4 cm (Fig. 1). The perimeter of the FL also had a recessed ledge lined with rubber to facilitate sealing of the tray to the bottom of the soil block. The FL was prepared by placing a circular sheet of glass wool over each drain spout in each funnel, adding loose glass wool on top of the filter, and then filling the funnels and border trays with clean pea-size gravel (1.7–2.4 mm diameter) until flush with the top of the funnels and trays.

Before each soil block was inserted into the stand, the bottom surface was lightly brushed with a wire brush to erase smeared surfaces and expose fresh pores. The bottom lid was then reattached, and the soil block lifted onto the stand by four people and later by a small front-end loader. The bottom lid on the soil block was then removed. The recessed ledge on the FL was coated with bees-wax, the FL attached to the bottom of the soil block and held in place with portable jack-stands, and the FL secured to the box with screws. A glass wool sheet was placed over the upper soil surface and a tracer-free solution (de-aired water containing 0.005 M CaSO_4 and 0.001 M thymol) was very slowly dripped onto the soil surface to saturate the soil. Thymol is a biological inhibitor. This slow drip was continued until one pore volume of water was added to the soil block, and then at least three pore volumes of water were leached through the soil to ensure saturation.

a) Top-view



B) Side-view

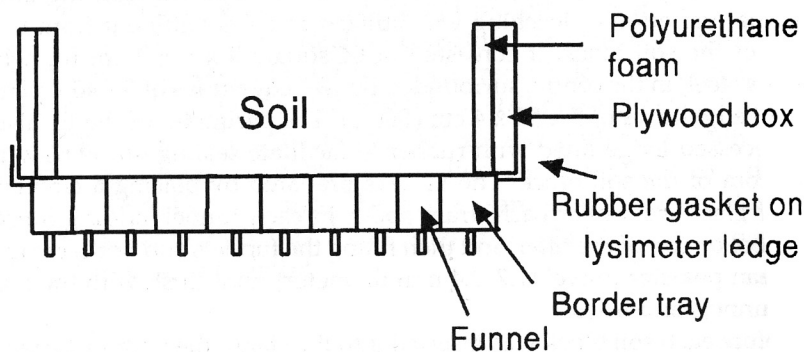


Fig. 1. Top-view (a) and side-view (b) of funnel-lysimeter used to collect effluent from bottom of large undisturbed soil blocks from southern Alberta.

After one pore volume of water had been initially added to the soil block, time-domain reflectometry (TDR) probes for monitoring soil moisture as outlined by Topp (1993) were installed at depths of 10, 25 and 45 cm below the upper soil surface. The TDR probes (three-probe configuration, stainless-steel rods, 34.2 cm length, 4.8 mm diam., 3.0 cm spacing) were installed so that about 29 cm of the rod extended in from the outside edge of the soil. The three probes were connected to a Tektronix Model 1502C cable tester with a coaxial cable, travel times recorded, and soil moisture

determined using the equation of Topp (1993). Soil moisture measurements were taken after four pore volumes of water had been leached through each soil block, and when TDR readings for all depths were constant, the miscible displacement experiment was initiated. TDR readings were also taken periodically during subsequent miscible displacement experiments.

A schematic diagram showing the experimental setup is shown in Fig. 2. About 1 to 2 cm of fine sand (0.13 mm diameter) was added to the soil surface to achieve a uniform hydraulic head of ponded water and to protect the soil surface from surface sealing during subsequent rainfall experiments. Before applying the sand, a glass-wool sheet was laid on the soil surface to prevent sand from entering and plugging macropores. A Mariotte bottle (50 L) was used to maintain a constant hydraulic head of 1 cm above the sand layer. A constant head of water was applied until steady-state conditions (inflow=outflow) were achieved.

Once steady-state conditions were achieved, a miscible displacement experiment was conducted. Application of the CaSO_4 solution was stopped and the head of water allowed to decline. When the tracer-free solution had just disappeared from the soil surface, a slug-type input (3150 ml) of tracer solution ($0.005\text{M CaCl}_2 \cdot \text{H}_2\text{O}$) was applied to the soil surface until a uniform head of 1 cm was achieved. The tracer solution was allowed to infiltrate, and when the tracer solution just disappeared from the soil surface, the tracer-free solution was reapplied until the end of the experiment. The volume of effluent exiting the entire funnel-lysimeter (funnels and border trays) per unit time (usually 10 minutes)

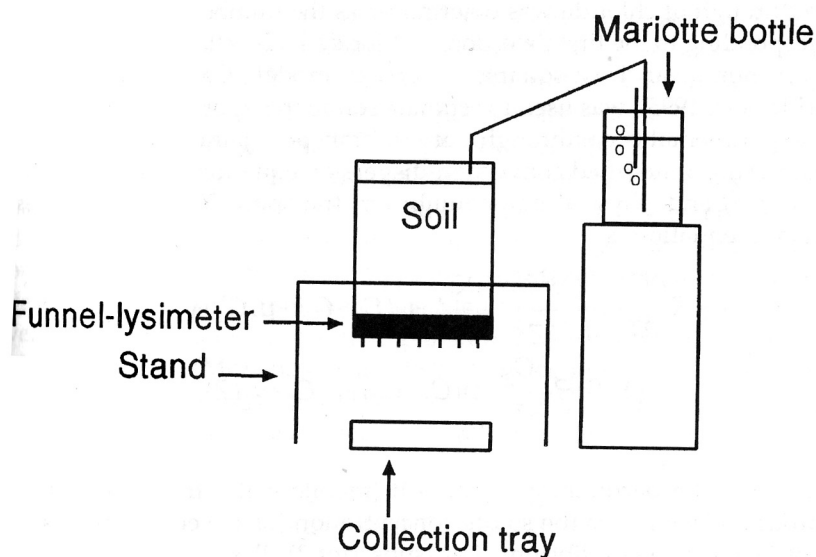


Fig. 2. Schematic of experimental setup for miscible displacement experiment.

was collected in a large stainless-steel pan every 10 minutes for the first 4–6 hours, and then every 0.5 and 1 hour, until about four pore volumes (250 L) of effluent had leached through the soil block. The mass of effluent was measured by weighing it on a balance to two decimal places and converting to volume, assuming a specific gravity of 1.0 g cm⁻³ for water. A subsample was collected, stored at 4°C, and analyzed for chloride ion concentration using the mercury thiocyanate-autoanalyzer method (Technicon 1974). The detection limit for chloride was 2.13 mg L⁻¹.

Determination of Breakthrough Curve Parameters

Breakthrough curves of relative chloride concentrations versus number of pore volumes eluted were derived from miscible displacement experiments. Relative chloride concentrations, c_0/c , (where c_0 is original chloride concentration in tracer solution, 354.5 mg L⁻¹, and c is chloride concentration at time t) were plotted as a function of number of pore volumes ($T=vt/L$, where T is dimensionless) calculated from

$$T = \frac{Qt}{A\theta L} \quad (1)$$

where Q is volumetric discharge of effluent per unit time (L³ T⁻¹), A is the cross-sectional area of flow (L²), θ is the volumetric water content or water-filled porosity (L³ L⁻³), T is the cumulative time (T) since the addition of tracer, and L is the vertical length (L) of the soil block. The value of one pore volume (water capacity) for each soil block was calculated by multiplying the volumetric water content by the total volume of the soil block. Initial breakthrough of chloride was determined as the number of pore volumes corresponding to the first detection of chloride in the effluent.

A nonlinear least-squares inversion model, CXTFIT-Version 2.0 (Toride et al. 1995), was used to estimate solute transport parameters from the experimental breakthrough curves. Transport parameters were estimated using a modified convection-dispersion equation, which describes two-region and physical non-equilibrium transport. The dimensionless transport equation is

$$\begin{aligned} \beta R \frac{\partial C_1}{\partial T} &= \frac{1}{P} \frac{\partial^2 C_1}{\partial Z^2} - \frac{\partial C_1}{\partial Z} - \omega(C_1 - C_2) - \mu_1 C_1 + \gamma_1(Z) \\ (1 - \beta)R \frac{\partial C_2}{\partial T} &= \omega(C_1 - C_2) - \mu_2 C_2 + \gamma_2(Z) \end{aligned} \quad (2)$$

where β is the partitioning coefficient (mobile water fraction), R is the retardation factor, C is the solute concentration for the equilibrium (subscript 1) and non-equilibrium sites (subscript 2), T is the number of pore volumes, P is the Peclet number, Z (x/L) is the ratio of the distance (x) to the column length (L), ω is the dimensionless mass transfer coefficient, μ

is the first-order rate constant for degradation of the solute, and γ is the zero-order rate constant. The model assumes that ω and μ cannot be negative. In CXTFIT 2.0, the pore-water velocity, v , (q/θ , where q is Darcy flow rate and θ is the volumetric moisture content), and dispersion coefficient, D , are used as input parameters instead of v_m and D_m for the mobile phase (m). The Darcy flow rate was derived from the saturated hydraulic conductivity (K_{sat}), assuming a unit hydraulic gradient. For the two-region model, β represents the mobile water fraction (θ_m/θ , where θ_m is the mobile water content and θ is the total water content), which we use as an estimate of preferential flow.

Flux-averaged concentrations were used for the concentration mode in the model. The upper boundary value problem was solved using a Dirac Delta input condition (instantaneous application of solute), using a relative chloride concentration of one. The invoked outlet boundary condition assumes a relative concentration of zero at an infinite distance. The initial value problem was solved using a zero initial concentration. For fitting of the model to the experimental breakthrough curves, the known input parameter was v , and the model best fit provided estimates for D , R , β , and ω .

Four K_{sat} tests were conducted during the miscible displacement experiment by separately collecting the volume of effluent per unit time in each funnel and the border trays. A tray of 64 collection bottles was inserted under the soil block to collect the funnel effluent, and eight 2-L plastic containers (2 per side) were used to collect effluent from border trays. Mean K_{sat} values for the whole soil block (border trays excluded) were calculated. The number of visible ($>500\ \mu\text{m}$) macropores on the bottom of each soil block was determined immediately before the miscible displacement experiment. Soil bulk density was determined by taking undisturbed soil cores (3 cm long $\times$ 5.5 cm diam.) from soil adjacent to each block in the field. Duplicate cores were taken at 5-cm depth increments from 0 to 50 cm. Bulk density was determined from the oven-dry weight and volume of the soil core. Bulk density values for each depth were averaged to obtain a mean value for each soil block.

In this experiment there was no replication of the CT and NT tillage treatments. Consequently, means and standard errors are reported for tillage fields to indicate overall trends, but statistical probability levels are not included. Ideally, the most accurate assessment of tillage treatment effects could be made from well-replicated field plot experiments; however, useful information can still be gained from long-term non-replicated field experiments (McKenzie et al. 1992). Long-term non-replicated plots are particularly useful for the detection of trends, and these hypotheses can then be more rigorously tested on replicated plots.

Results and Discussion

Selected physical soil properties and hydraulic parameters, some of which were used as input parameters for the model (e.g., block length,

pore-water velocity), are shown on Table 1. Mean bulk density and number of visible pores were higher for the CT field than the NT field, whereas volumetric moisture content, Darcy flow rate, and pore-water velocity were relatively similar.

In general, reasonably good fits were obtained for the two-region model to the observed breakthrough curves (BTC). Our correlation coefficients, however, were lower (0.60–0.84) than those ($r^2 > 0.967$) reported by Jensen et al. (1996), who also fitted a two-region model to BTC of chloride in undisturbed soil (loamy sand). Breakthrough curves (BTC) for both tillage fields exhibited early initial breakthrough, a rapid rise in tracer concentration, a shift of the BTC peak to the left of one pore volume, and a slow decline ("tailing") toward zero tracer concentration for the descending limb of the curve (Fig. 3 and 4). Tailing has been attributed to physical processes involving mobile-immobile regions (van Genuchten and Wierenga 1976) or to chemical processes involving two-site adsorption (Cameron and Klute 1977). Mean values for initial breakthrough of chloride in terms of pore volumes were similar for the CT field (0.12) and the NT field (0.09). Singh and Kanwar (1991) reported similar pore volume values for initial breakthrough of chloride for CT (0.13) and NT (0.11) soils in Iowa. The time of initial breakthrough is an important parameter which is closely associated with soil purification of liquid waste, and soils that exhibit early initial breakthrough do not act as very effective filters for liquid wastes (Anderson and Bouma 1977).

Table 1. Selected physical properties and hydraulic parameters of undisturbed clay loam soil blocks taken from long-term (since 1968) conventional tillage (CT) and no-till (NT) fields in southern Alberta

Treatment	Bulk density	No. visible pores	Vol. water content (θ)	Block length (L)	Darcy flow rate, q ($q=K_{sat}$)	Pore-water velocity, v ($v=q/\theta$)
	g cm^{-3}		$\text{cm}^3 \text{ cm}^{-3}$	cm	cm day^{-1}	cm day^{-1}
CT-1	1.36	28	0.454	50	210.0	462.6
CT-2	1.37	355	0.455	50	167.6	368.4
CT-3	1.40	583	0.454	50	128.7	283.5
CT-4	1.37	128	0.469	50	159.8	340.7
Mean $\pm$ SE	1.38 $\pm$ 0.009	274 $\pm$ 124	0.458 $\pm$ 0.004	—	166.5 $\pm$ 16.8	363.8 $\pm$ 37.4
NT-1	1.27	54	0.438	52	178.9	408.4
NT-2	1.31	13	0.453	49	117.5	259.4
NT-3	1.36	205	0.435	50	112.3	258.2
NT-4	1.33	160	0.458	51	262.7	573.6
Mean $\pm$ SE	1.32 $\pm$ 0.02	108 $\pm$ 45	0.446 $\pm$ 0.006	—	167.9 $\pm$ 35.0	374.9 $\pm$ 75.0

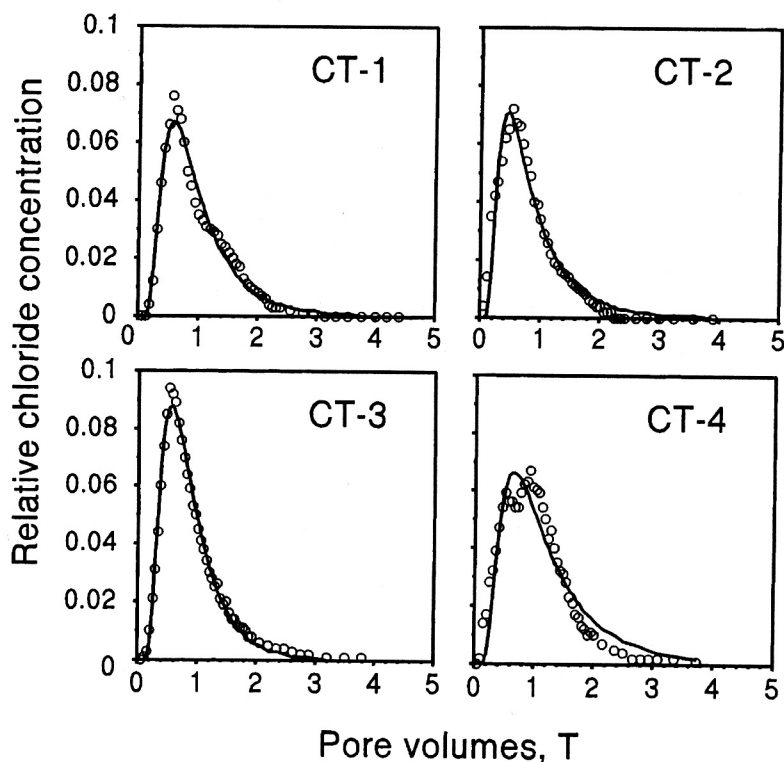


Fig. 3. Breakthrough curves from four large undisturbed soil blocks taken from conventional tillage (CT) field in southern Alberta. Open circles represent measured values, solid line is best fit from model simulation.

Mean dispersion coefficients (D) were slightly higher for the NT field ($1.80 \text{ cm}^2 \text{ day}^{-1}$) than the CT field ($1.40 \text{ cm}^2 \text{ day}^{-1}$) (Table 2). The dispersion coefficient includes the effects of both mechanical dispersion, D_m , and diffusive transport, D_h (van Genuchten and Wierenga 1986). Dispersive transport is caused by deviation of local fluid velocities from the average pore-water velocity inside individual pores and between pores of different shapes, sizes and directions (van Genuchten and Wierenga 1986). Dispersion is a passive process that, unlike diffusion, occurs only during water movement. In comparison, Andreini and Steenhuis (1990) found slightly higher D values than our study, with higher mean D values for NT ($13.1 \text{ cm}^2 \text{ day}^{-1}$) than CT ($9.7 \text{ cm}^2 \text{ day}^{-1}$). Singh and Kanwar (1991) reported considerably higher D values than us, with higher D values for NT ($3456 \text{ cm}^2 \text{ day}^{-1}$) than CT ($13824 \text{ cm}^2 \text{ h}^{-1}$). In a review of the literature, Bevan et al. (1993) reported a range in D values from 3.9 to $8880 \text{ cm}^2 \text{ day}^{-1}$ for various undisturbed soil cores.

Mean values for the retardation factor (R) were higher for the NT field (0.42) than the CT field (0.32) (Table 2). Retardation factors less than

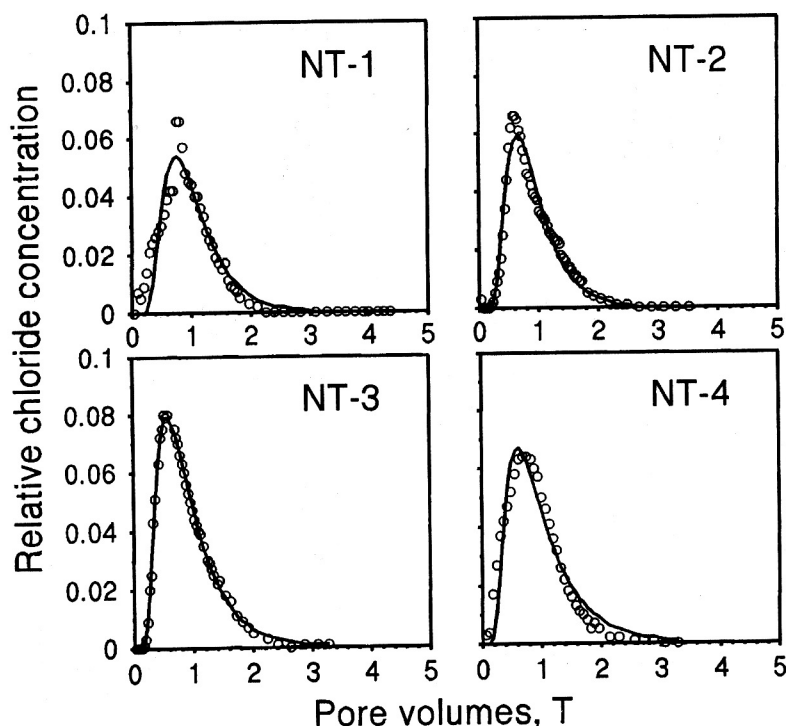


Fig 4. Breakthrough curves from four large undisturbed soil blocks taken from no-till (NT) field in southern Alberta. Open circles represent measured values, solid line is best fit from model simulation.

one indicate that only a fraction of the liquid phase participates in the transport process, which may be due to anion exclusion or immobile water regions (Wierenga and van Genuchten 1989). In comparison, Singh and Kanwar (1991) reported higher R values for CT (0.51) than NT (0.44). Our range of R values (0.25–0.50) was lower than other values reported for miscible displacement of chloride in the literature (Wierenga and van Genuchten 1989; Jensen et al. 1996). This trend may be due to greater anion exclusion in our soils. Clay and organic matter are the main contributors to anion exclusion in soils (Bolt 1979), and the clay content in our soils (28–36%) was considerably higher than values (4.7–8%) reported by others (Wierenga and van Genuchten 1989; Jensen et al. 1996).

Mean values for the mobile water fraction (β) were similar for CT (0.78) and NT (0.80), suggesting no difference in preferential leaching of chloride (Table 2). Singh and Kanwar (1991) estimated the mobile pore-water fraction directly from their BTCs as the number of pore volumes required to reach a relative chloride concentration of 0.5, and the immobile fraction as one pore volume minus the mobile water fraction. They found that the immobile pore-water fraction was higher for NT (56%)

Table 2. Optimized parameters for fitting of two-region flow model to break-through curves of undisturbed clay loam soil blocks taken from long-term (since 1968) conventional tillage (CT) and no-till (NT) fields in southern Alberta

Treatment	Dispersion coef. (D)	Retardation factor (R)	Mobile water fraction (β)	Mass transfer coeff. (ω)	Initial break- through chloride	Correl. coeff. (r^2)
	cm ² day ⁻¹					
CT-1	2.72	0.42	0.92	0.05	0.28	0.77
CT-2	0.51	0.25	0.81	0.06	0.05	0.84
CT-3	2.08	0.25	0.65	0.09	0.07	0.70
CT-4	0.29	0.37	0.73	0.09	0.07	0.76
Mean $\pm$ SE	1.40 $\pm$ 0.59	0.32 $\pm$ 0.04	0.78 $\pm$ 0.06	0.07 $\pm$ 0.009	0.12 $\pm$ 0.05	—
NT-1	0.26	0.41	0.82	0.07	0.08	0.68
NT-2	1.81	0.30	0.78	0.06	0.13	0.78
NT-3	0.50	0.46	0.82	0.09	0.11	0.71
NT-4	4.64	0.50	0.77	0.09	0.05	0.60
Mean $\pm$ SE	1.80 $\pm$ 1.00	0.42 $\pm$ 0.04	0.80 $\pm$ 0.01	0.08 $\pm$ 0.006	0.09 $\pm$ 0.02	—

than CT (49%), which indicated greater preferential flow in NT. Jensen et al. (1996) reported β values ranging from 0.28 to 0.89 (mean $\pm$ SD; 0.64 $\pm$ 0.20) for saturated miscible displacement of chloride through undisturbed soil cores (loamy sand, 1 x .2 m diam cores) in Denmark. Li and Ghodrati (1994) found that β values ranged from 0.21 to 0.90 for miscible displacement of NO₃ through old root channels (corn, alfalfa) in a silt loam soil (repacked columns). Beven et al. (1993) reviewed the literature on dispersion parameters in undisturbed, partially saturated soil cores, and reported β values ranging from 0.32 to 1.0.

We expected that preferential flow would have been higher for the NT than CT field because of the high populations of earthworms (*Aporrectodea caliginosa*) found in the NT field, and the absence of earthworms in the CT field (Clapperton et al. 1997). At the time of sampling, earthworms were found in soil adjacent to blocks NT-1, NT-3 and NT-4, but none were found adjacent to the other blocks. Earthworms were found in block NT-1 between the 0 and 30 cm depth, with maximum densities (48 m²) at the 5 to 10 and 20 to 25 cm depths. In block NT-3, they were found between the 0 and 15 cm depth, with maximum densities (208 m²) at the 0 to 5 cm depth. In block NT-4, they were found between the 0 and 25 cm depth, with maximum densities (272 m²) at the 5 to 10 cm depth. Previous studies in these fields have reported higher K_{sat} values (small soil cores) for the NT field (0 to 20 cm depth) than the CT field,

which was attributed to earthworm burrows (Miller et al. 1999). We believe that the concentration of earthworms at a shallow depth (<30 cm) in our NT blocks may have resulted in little difference in preferential leaching at the 50 cm depth.

Old root channels may have exerted a strong influence on solute leaching in these soil blocks, since soil blocks were stored for a relatively long period of time. We observed considerably more old root channels (>500 μm diam.) on the bottom of the CT soil blocks (274) than the NT (108) blocks (Table 1). Volkmar and Entz (1995) found greater rooting density in the CT field than the NT field, and attributed this trend to greater penetration resistance and compaction under NT. However, we found no correlation between visible number of pores and the mobile water fraction (β) for each soil block. The lack of close correspondence between solute leaching and visible pores has been previously observed, and has been attributed to the inability to estimate macropore continuity from two-dimensional sections (Singh et al. 1991; Trojan and Linden 1992; Granovsky et al. 1993).

Soil structure in both tillage fields was similar (medium to very coarse subangular blocky primary structure in A horizon and medium to very coarse prismatic primary structure in B horizon), which suggested little potential influence of soil peds or adjacent cracks on solute leaching. Lauren et al. (1988) have suggested that for an undisturbed soil block to accurately represent saturated water flow in the field, the block should sample a representative elementary volume (REV) of at least 20 peds. An REV is the smallest volume of soil that contains a representation of microscopic variations in all the forms and proportions present in the system (Bear 1972). Assuming a subangular ped size of 125 cm^3 from 0 to 20 cm depth (50,000 cm^3) gives an REV of 400 peds, and assuming a prismatic ped size of 4,000 cm^3 from 20 to 50 cm depth (75,000 cm^3), gives a REV of 19 peds. Therefore, the total REV for our soil blocks (108,000 cm^3) was approximately 419 peds, which met and greatly exceeded the minimum REV guideline of 20 peds. Lauren et al. (1988) conducted K_{sat} tests on undisturbed soil blocks of similar texture (silty clay loam) and structure (subangular blocky with medium prismatic), and found that a sample volume of 50,000 cm^3 (50 $\times$ 50 $\times$ 20 cm) was an optimum size that minimized CV values and met the REV criterion.

No tillage effect for preferential leaching of chloride may have been related to the timing of sampling in relation to the crop-fallow sequence and the last tillage operation for CT as well as the long period of storage for the soil blocks. We sampled the soil blocks during the fallow phase, which may have affected preferential leaching. Chang and Lindwall (1989) found that steady infiltration rates were significantly higher for NT than CT under fallow, but there was no significant difference under fresh stubble. Since soil blocks in the CT field were taken at least 21 days after the most recent tillage event, and soil blocks were stored for a considerable period (32 months) without tillage, the surface soil of CT blocks may have consolidated, resulting in reduced infiltration and leaching.

Previous researchers have generally reported few significant differences in infiltration between CT and NT when infiltration measurements are made several weeks after the last tillage operation (Maulé and Reed 1993). In contrast, when infiltration measurements are made immediately after a tillage operation, the CT system usually results in higher infiltration rates than NT because of greater porosity and surface detention (Maulé and Reed 1993).

Short-term tillage effects on soil physical properties caused by tillage may be masked after as little as 4 weeks after the last tillage operation (Chang and Lindwall 1989). Ideally, soil physical measurements should be taken immediately after each management or weather event (i.e., cultivation or intense rainfall) because of large temporal variations in soil physical properties under different tillage systems (Logsdon et al. 1993). However, this is often not possible in practice, so we recommend that the timing of measurements in relation to sampling and tillage events as well as storage time be reported, and the results be presented in this context. Because of the long storage period for our soil blocks, we focussed on the long-term (e.g., soil structure, rooting density, earthworm burrows) rather than short-term (e.g., tillage of surface soil) effects of tillage practice on preferential leaching.

We believe our results for relative differences between CT and NT are more applicable to scenarios where considerable time has elapsed between the last tillage operation for CT and the next major rainfall or leaching event. Caution is advised in extrapolating our results to scenarios of significant rainfall or leaching events immediately after plowing. Andreini and Steenhuis (1990) determined breakthrough curves for CT and NT, and noted that their relative results between CT and NT may have been different had their soil cores been taken a shorter time after the CT soil was tilled. Their soil cores were also stored for 12 months before being tested. Further research is needed to compare leaching between CT and NT for the scenario of a major rainfall event immediately after tillage.

Summary and Conclusions

Breakthrough curves for both tillage fields exhibited early initial breakthrough, a rapid rise in tracer concentration, a shift of the BTC peak to the left of one pore volume, and a slow decline ("tailing") toward zero concentration for the descending limb of the curve. Reasonably good fits were obtained for fitting of the two-region model to the BTCs, as indicated by correlation coefficients ranging from 0.60 to 0.84. Mean values for the mobile water fraction were similar for the CT field (0.78) and NT field (0.80), suggesting no difference in preferential leaching of chloride. We hypothesize that the extent of preferential leaching in earthworm burrows (*Aporrectodea caliginosa*) in the NT field and leaching in old root channels in the CT field may have been similar. Further research on replicated plots is needed to examine short-term effects of tillage practice (e.g.,

tillage of surface soil) as well as the individual contribution of earthworm burrows and old root channels to preferential leaching under CT and NT. In addition, additional study is required on the effect of tillage practice on the extent of preferential leaching under unsaturated conditions.

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